ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

NAME:NARAPPAGARI SRAVYA

REGISTRATION NUMBER:192472192

PROGRAMME:B.E CSE - AI

SLOT:D COURSE CODE:DSA0305

COURSE NAME:NATURAL LANGUAGE PROCESSING

COURSE FACULTY: Dr.K.ANBAZHAGAN

Section 1: Resume Information Extraction System

CODE:

```
import re

resumes = [

"""

Name: Rahul Sharma

Email: rahul.sharma@gmail.com

Mobile: 9876543210

Skills: Python, Java, SQL, Machine Learning

Experience: 3 years

""",


"""

Name: Priya Reddy

Email: priya.reddy@yahoo.com

Mobile: 9123456789

Skills: Java, SQL, NLP

Experience: 1 years

""",


"""
```

```
Name: Arun Kumar
Email: arun123@gmail.com
Mobile: 9988776655
Skills: Python, NLP
Experience: 5 years
"""
]
skills_list = ["Python", "Java", "SQL", "Machine Learning", "NLP"]

print("========= Resume Information =========")

eligible = []

for resume in resumes:

    name = re.search(r"Name:\s*(.*)", resume)
    email = re.search(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", resume)
    mobile = re.search(r"\b[6-9]\d{9}\b", resume)
    experience = re.search(r"(\d+)\s+years", resume)

    skills = []
    for skill in skills_list:
        if re.search(skill, resume, re.IGNORECASE):
            skills.append(skill)

    print("\nCandidate Summary")
    print("------------------------")
    print("Name:", name.group(1))
    print("Email:", email.group())
```

```
    print("Mobile:", mobile.group())
    print("Skills:", ", ".join(skills))
    print("Experience:", experience.group(1), "years")

    if int(experience.group(1)) >= 2 and "Python" in skills:
        eligible.append(name.group(1))

print("\nEligible Candidates")
print("-------------------")
for candidate in eligible:
    print(candidate)
```

OUTPUT:

```
========== Resume Information ==========

Candidate Summary
-------------------------
Name: Rahul Sharma
Email: rahul.sharma@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning
Experience: 3 years

Candidate Summary
-------------------------
Name: Priya Reddy
Email: priya.reddy@yahoo.com
Mobile: 9123456789
Skills: Java, SQL, NLP
Experience: 1 years
```

Candidate Summary

Name: Arun Kumar

Email: arun123@gmail.com

Mobile: 9988776655

Skills: Python, NLP

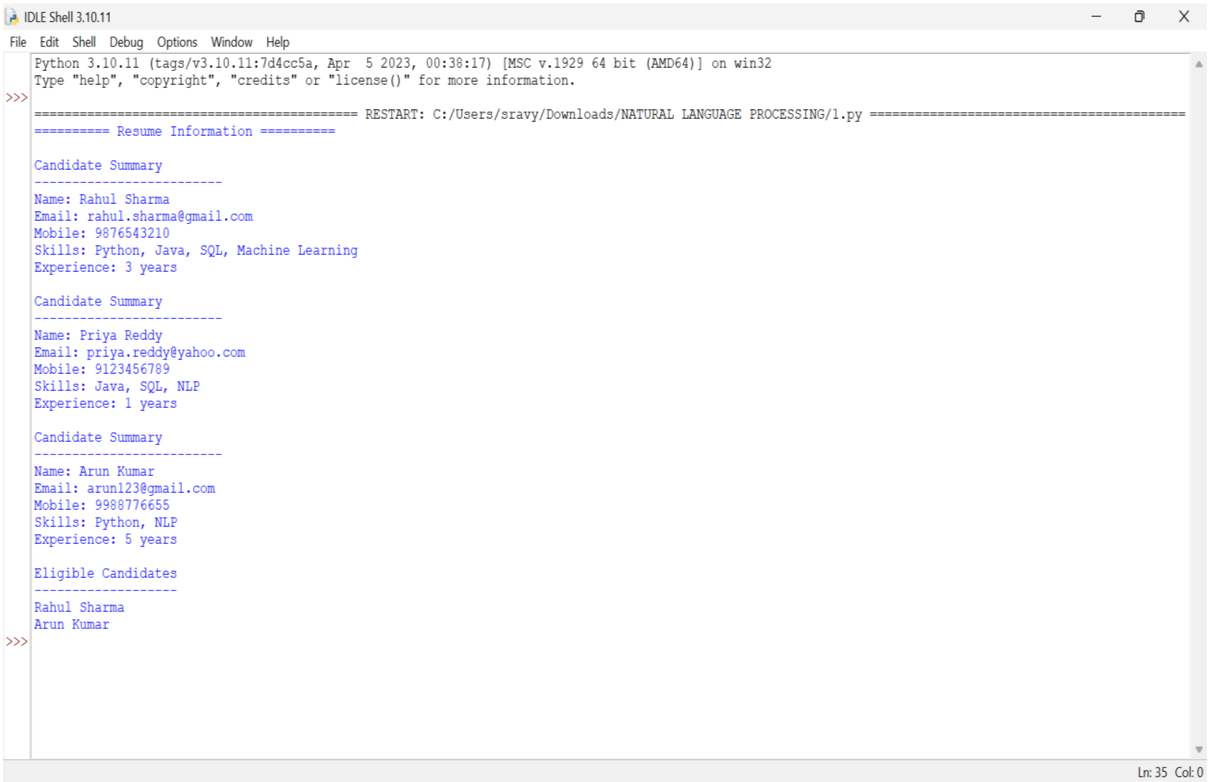
Experience: 5 years

Eligible Candidates

Rahul Sharma

Arun Kumar

=== Code Execution Successful ===



Section 2: Product Search System

CODE:

```
import re


products = [
    "Laptop",
    "Laptop Bag",
    "Gaming Laptop",
    "Wireless Mouse",
    "Bluetooth Speaker",
    "Smart Phone",
    "Phone Cover",
    "Python Programming Book",
    "Java Programming Book",
    "SQL Guide"
]


def search(keyword):


    print("\nExact Match")
    exact = [p for p in products if re.fullmatch(keyword, p, re.IGNORECASE)]
```

```
    print(exact)

    print("\nPrefix Search")
    prefix = [p for p in products if re.search("^" + keyword, p, re.IGNORECASE)]
    print(prefix)

    print("\nSuffix Search")
    suffix = [p for p in products if re.search(keyword + "$", p, re.IGNORECASE)]
    print(suffix)

    print("\nPartial Search")
    partial = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
    print(partial)

    print("\nCase-Insensitive Search")
    case = [p for p in products if re.search(keyword, p, re.IGNORECASE)]
    print(case)

    print("\nReport")
    print("Exact Matches :", len(exact))
    print("Prefix Matches:", len(prefix))
    print("Suffix Matches:", len(suffix))
    print("Partial Matches:", len(partial))

keyword = input("Enter search keyword: ")
search(keyword)
```

OUTPUT:

Enter search keyword: LAPTOP

Exact Match

['Laptop']

Prefix Search

['Laptop', 'Laptop Bag']

Suffix Search

['Laptop', 'Gaming Laptop']

Partial Search

['Laptop', 'Laptop Bag', 'Gaming Laptop']

Case-Insensitive Search

['Laptop', 'Laptop Bag', 'Gaming Laptop']

Report

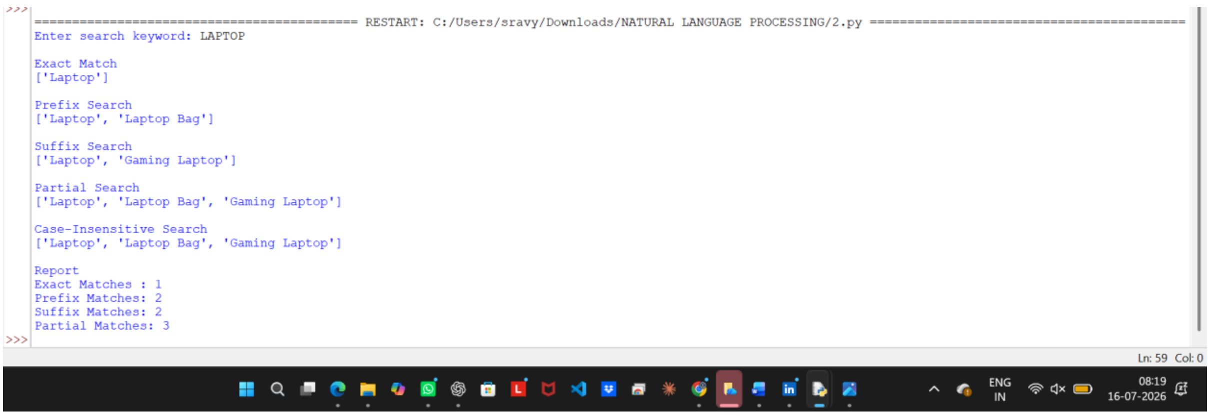
Exact Matches : 1

Prefix Matches: 2

Suffix Matches: 2

Partial Matches: 3

=== Code Execution Successful ===



Section 3: University Registration Validation System

CODE:

```
import re


reg_no = input("Enter Register Number: ")

email = input("Enter Institution Email: ")

course = input("Enter Course Code: ")

semester = input("Enter Semester: ")

mobile = input("Enter Mobile Number: ")


status = True


# Register Number (Example: 22CSE001)

if re.fullmatch(r"\d{2}[A-Z]{3}\d{3}", reg_no):

    print("Register Number: Valid")

else:

    print("Register Number: Invalid")

    status = False


# Institution Email

if re.fullmatch(r"[a-zA-Z0-9._%+-]+@college\.edu", email):
```

```
    print("Email: Valid")
else:
    print("Email: Invalid")
    status = False

# Course Code (Example: CS101)
if re.fullmatch(r"[A-Z]{2,3}\d{3}", course):
    print("Course Code: Valid")
else:
    print("Course Code: Invalid")
    status = False

# Semester (1-8)
if re.fullmatch(r"[1-8]", semester):
    print("Semester: Valid")
else:
    print("Semester: Invalid")
    status = False

# Mobile Number
if re.fullmatch(r"[6-9]\d{9}", mobile):
    print("Mobile Number: Valid")
else:
    print("Mobile Number: Invalid")
    status = False

print("\n========== Registration Report ==========")


if status:
```

```
    print("Registration Successful")
else:
    print("Registration Failed")
```

OUTPUT:

Enter Register Number: 23ECE205

Enter Institution Email: student@college.edu

Enter Course Code: EC205

Enter Semester: 3

Enter Mobile Number: 8767654321

Register Number: Valid

Email: Valid

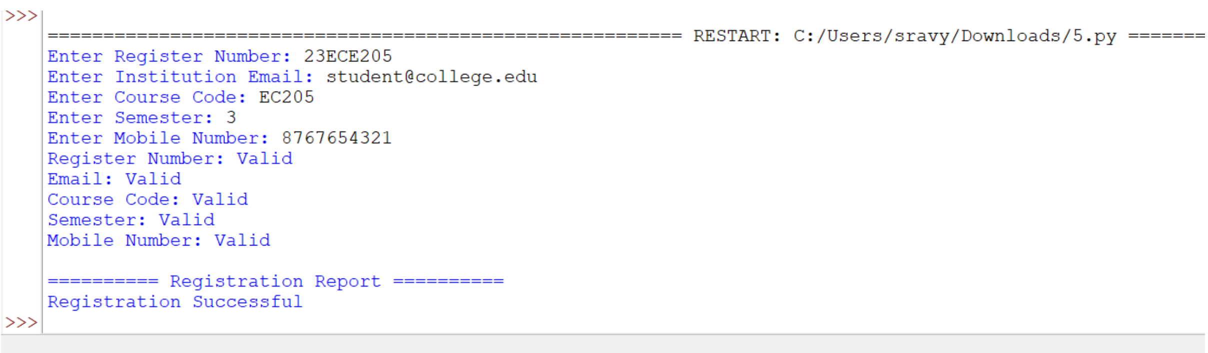
Course Code: Valid

Semester: Valid

Mobile Number: Valid

========== Registration Report ==========

Registration Successful



```
>>>
======================================================= RESTART: C:/Users/sravy/Downloads/5.py =======
Enter Register Number: 23ECE205
Enter Institution Email: student@college.edu
Enter Course Code: EC205
Enter Semester: 3
Enter Mobile Number: 8767654321
Register Number: Valid
Email: Valid
Course Code: Valid
Semester: Valid
Mobile Number: Valid

========== Registration Report ==========
Registration Successful
>>>
```